

1. Introduction

Mineral exploration targeting, which involves identifying and prioritizing locations most likely to contain economically viable mineral deposits, is a decision problem characterized by high impact and high uncertainty (Carranza, 2009; Haldar, 2013; Hronsky & Groves, 2008; Nykänen et al., 2025; Partington et al., 2024, 2024; Singer, 1993; Yousefi et al., 2024). Targeting decisions can lead to actions, such as acquiring more data or drilling, that require significant capital investments. These early decisions are especially consequential because fewer than 1% to 5% of explored prospects advance to economically viable mines (McIntosh, 2016; Regueiro González-Barros & Espí, 2019). Poor targeting can lead to costly drilling or data acquisition in areas with limited economic potential. Initial targeting, therefore, represents a critical, high-leverage decision point within the mining life cycle.

Traditionally, the mineral exploration industry manages targeting risk and high uncertainty through an expert-led and multiscale workflow. Gradually narrow the search from a regional to a target scale by utilizing larger volumes of geophysical, geological, and geochemical data to detect anomalies (Carranza, 2009; Hronsky & Groves, 2008; McCuaig et al., 2010). This is then followed by testing these anomalies with increasingly expensive methods. Recently, data-driven methods have been gaining popularity. They can augment expert interpretation by extracting meaningful targeting information from large, multi-source datasets. They provide more systematic, less biased, and more reproducible exploration workflows (Yang et al., 2024; Yousefi et al., 2021).

Current work in data-driven mineral exploration largely focuses on mineral prospectivity mapping. This approach combines multiple exploration evidence layers to predict favorable locations for mineralization (Carranza, 2009; Yousefi et al., 2021; Zuo, 2020). Prospectivity-mapping methods range from classical GIS-based evidence integration to newer machine learning and AI models (Yang et al., 2024; Zuo, 2020). Exploratory multivariate methods are widely used to summarize large geoscience datasets for mineral targeting. PCA and factor analysis are commonly used to reduce dimensionality and redundancy among correlated geochemical, geophysical, and remote-sensing variables. By transforming the original variables into a smaller set of components or latent factors, these methods reveal dominant patterns of covariation. They provide a reduced feature space in which samples or map cells with similar multivariate signatures can be compared or grouped. (Jansson et al., 2022; Sadeghi et al., 2024; J. Wang et al., 2017; L. Wang et al., 2024). Nonlinear embedding and representation-learning methods, such as t-SNE, UMAP, and autoencoders, can capture complex nonlinear relationships in high-dimensional exploration datasets that may not be well represented by linear methods such as PCA or factor analysis. They can support exploratory visualization and pattern recognition. This includes the identification of clusters or gradients related to lithology, alteration, or mineralization (Luo et al., 2020; McInnes et al., 2018; van der Maaten & Hinton, n.d.; Zhang et al., 2024).

Although complementary data-driven methods can support expert interpretation, they can be extended in two important ways. First, methods can be designed to work with fewer labeled

examples. This is important because, in real exploration settings, only a few deposits may be known for a given mineral system (Chen & Wu, 2017; Xiong & Zuo, 2021). Second, known deposits can be used as spatial geophysical signatures, rather than only as labeled locations. Their geophysical geometry can contain important information for identifying analogous mineralization (Yu et al., 2024). Together, these opportunities motivate methods that can work with few labeled examples and make better use of the spatial information contained in each known deposit, beyond its location alone.

Therefore, this paper aims to develop a method to learn the geometry of a single known deposit as expressed in multivariate geophysical data and to search for similar geometries across a study area. The known deposit area is constrained by existing drilling and resource assessment. The method aims to rank other areas based on how closely their geometry across multi-variate datasets matches the known deposit geometry.

We intend to show that this method introduces a quantitative metric of spatial similarity based on spatial principal component analysis (sPCA). By decomposing into spatial eigenfrequencies, sPCA captures the specific geophysical geometry of the known deposit and quantifies how closely other locations reproduce that geometry. We intend to show that sPCA produces spatial components that can be visualized, weighted according to the known deposit, and used directly to rank other areas by geometric similarity.

The aim is to enable the detection of deposit-specific geometry without labeled data or filters using multivariate geophysical datasets. Beyond being fully data-driven, the approach is intended to complement and enhance expert-driven exploration with an automated, fast, and scalable method.

We intend to show this approach in a brownfield case study in the Carajás Mineral Province (Brazil). We apply it to magnetic and radiometric datasets to identify IOCG-related geometries consistent with a known copper deposit.

2. Real Case Introduction

The study area lies in the Carajás Mineral Province of Pará, northern Brazil. This region hosts the Archean granite–greenstone and volcano-sedimentary terrane of the Itacaiúnas Supergroup. It

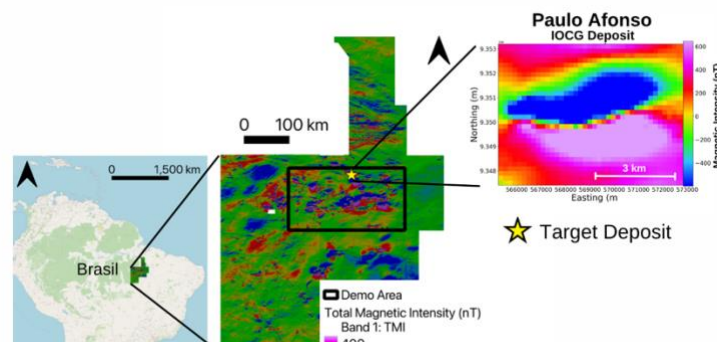


Figure 1. Location of study area.

contains numerous Fe-oxidized IOCG systems. We aim to detect locations within it whose geometry resembles that of a known IOCG deposit and rank them by similarity.

The training data for Case 1 is the Total Magnetic Intensity (TMI) in the Paulo Afonso deposit area. This is an iron-oxide copper–gold (IOCG) body characterized by magnetite–hematite–rich alteration and structurally focused chalcopyrite $\pm$ Au mineralization. This body produces a well-defined dipole geometry in the TMI grid. This dipole TMI response is a diagnostic geometry that IOCG experts already use as an exploration criterion in this area. This makes it a geologically meaningful template for identifying district-scale analogs and assessing regional prospectivity. By adopting this dipole as the reference geometry, our analysis is intentionally restricted to other areas that express a similar dipole geometry.

To enable a like-for-like comparison between the deposit and the surrounding areas, we analyze the TMI grid (Figure 1) using a sliding-window approach. Here, every window has exactly the same dimensions as the deposit footprint. The resulting analysis window measures 46×28 pixels (1,288 pixels in total). It corresponds to a physical area of 9.20×5.60 km at the 200-m grid resolution. We move this window across the full 259×138 km demonstration area ($2,485 \times 3,527$ pixels) using a fixed lateral and vertical stride. This extracts 13031 spatially consistent windows from the TMI grid. Each of which can be directly compared to the deposit's geophysical geometry.

Within this study area, several other economic IOCG and iron-ore deposits exhibit comparable magnetic dipoles (Tucuma, Pedra Branca, Salobo, etc.). These deposits provide an independent way to test whether the method identifies meaningful targets (Figure 2).

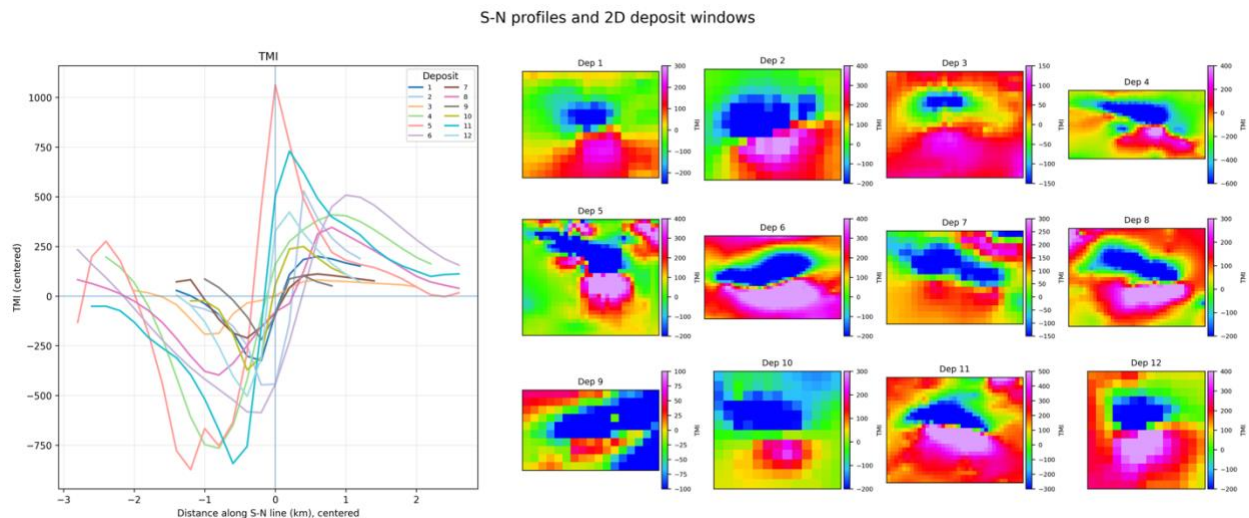


Figure 2. Centered south-north (S-N) TMI profiles and corresponding 2D deposit windows for the twelve deposits. The profiles summarize the 1D geometry, while the 2D windows show the full spatial patterns used in the spatial PCA analysis.

Case 2 focuses on Alemão (Deposit 3) as a representative multivariate example. In the Igarapé Bahia camp, IOCG mineralization also expresses a dipole geometry in the TMI grid. Alemão has been

described as a Cu-Au-(U-REE) IOCG deposit containing minor uraninite. Therefore, elevated radiometric U provides a geologically meaningful complementary signal in this area. The combined presence of a magnetic dipole and high radiometric U is treated here as a district-specific IOCG exploration signature. This motivates the multivariate TMI + U analysis developed in Case 2.

3. Methodology

The goal of the method is to create a data-driven similarity map and target ranking that shows which areas have geophysical geometries most similar to that of the known deposit. The methodology starts with a known deposit window constrained by drilling and resource assessment (**Error! Reference source not found.**). The geophysical data within this window define the reference geometry for comparison. We then move a window of the same size across the study area. Each sliding window is compared with the known deposit window. Windows with the closest match are ranked highest, under the assumption that similar geophysical geometry may indicate similar ore geometry. To compare the windows, we use spatial PCA. This decomposes the sliding-window data into orthogonal spatial components. Each window is represented by its PCA scores. We then compute the distance between each window and the known deposit window. We give more weight to the components most characteristic of the deposit geometry. This produces a similarity map that ranks windows by how closely they match the reference geometry.